

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. EE Sem - IV Examination, 2010-11

EE - 204 Electrical Power Systems-I

Date: 11/05/2011, Wednesday Time: 10 to 1 p.m Maximum Marks: 70

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate full marks.
4. Assume suitable additional data, if necessary

SECTION - I

- Q-1 A. What is electric power supply scheme? Draw & explain single line diagram of a typical a.c power supply scheme & Discuss advantage of high transmission voltage. 6
- B. Define and explain string efficiency. Can its value be equal to 100%? 2
- C. Determine the voltage across each disc of suspension insulator as a percentage of the line voltage to earth. The self and capacitance to ground of each disc is C and 0.2C respectively. The capacitance between the link pin and the guard ring is 0.1C. Determine the distribution of voltage in terms of % line voltage to earth, also determine efficiency 4
- Q-2 A. Discuss the various methods of determining the depreciation of the equipment. 5
- B. Write short note on two part tariff, power factor tariff, three part tariff 2
- C. A generating station has the following data:
Installed capacity=300MW; Capacity factor=50%; Annual load factor=60%; Annual cost of fuel, oil etc. =Rs 9×10^7 ; capital cost = Rs 10^9 ; annual interest & depreciation=10%.(i) The minimum reserve capacity of the station and (ii) Calculate the cost per kWh generated 4
- OR
- Q-2 A. Derive the equation of the optimum power factor. 5
- B. Explain the disadvantages and causes of low power factor. 2
- C. A factory operates at 0.8 pf lagging and has a monthly demand of 750kVA. The monthly power rate is Rs 8.50 per kVA. To improve the power factor 250kVAR capacitor are installed in which there is negligible power loss. The installed cost of equipment is Rs.20000 and fixed charges are estimated at 10% per year. Calculate the annual saving effected by the use of capacitors 4
- Q-3 A. Discuss the advantage of interconnected grid system. 3

- B. Explain briefly various types of conductors used in overhead transmission lines. 4
- C. What are the various methods of improving string efficiency? Discuss the advantage and disadvantage of
(1) Pin type insulator, (2) suspension type insulator 5

OR

- Q-3 A. Describe briefly commonly used insulating materials for cables. Compare the merits & demerits of underground system versus overhead system. 7
- B. A 33 kV, 50 Hz, 3- phase underground cable 4 km long uses three single core cables. Each of the conductors has a diameter of 2.5 cm and the radial thickness of insulation is 0.5 cm. Determine (i) capacitance of the cable/phase (ii) charging current/phase (iii) total charging kVAR. The relative permittivity of insulation is 3. 5

SECTION – II

- Q-4 A. Explain Ferranti effect with suitable vector diagram. 7
- B. Define: Surge impedance and surge impedance loading 3
- C. Define: Short and medium transmission line 2
- Q-5 A. The ABCD constants of one of a 3 phase transmission line are given as follows: $A=D=0.9+j0.012$, $B=22.5+j150\Omega$, $C=(-0.00004+j0.001)\text{mho}$. The sending end voltage is 240KV and the receiving end voltage is 220 KV (line to line voltage). Draw the circle diagram and determine the active power and reactive power received when the load angle between sending end and receiving end voltage phasors is 30 degree. 7
- B. Draw Π model and vector diagram of a medium transmission line. 4

OR

- Q-5 A. Derive the equation of receiving end active and reactive power in terms of ABCD constants of the transmission line. 7
- B. Explain Π - equivalent model of a long transmission line. 4
- Q-6 A. A 3- Φ double circuit line is shown in Fig.1. Radius of each conductor is 0.9 cm. Find the inductance per phase per km of the line. 7
- B. A single phase double circuit line is shown in Fig. 2. Conductors 1 and 2 in parallel from one path. Conductors 1' and 2' in parallel from return path. Determine the total inductance per km of the line. GMR of each conductor is $(0.7788 \times 10^{-2})\text{mt}$. 3
- C. Why is transmission lines transposed? 2

OR

- Q-6 A. Six conductors of a double circuit line are arranged as shown in Fig. 3. The diameter of each conductor is 2.5 cm. Find the capacitive reactance to neutral and the charging current per km per phase at 132KV. Neglect the effect of earth. 7
- B. Derive the equation of capacitance of 3 phase overhead transmission line with symmetrical spacing. 3
- C. Write equation of capacitance of 1- Φ transmission line considering the effect of earth. 2

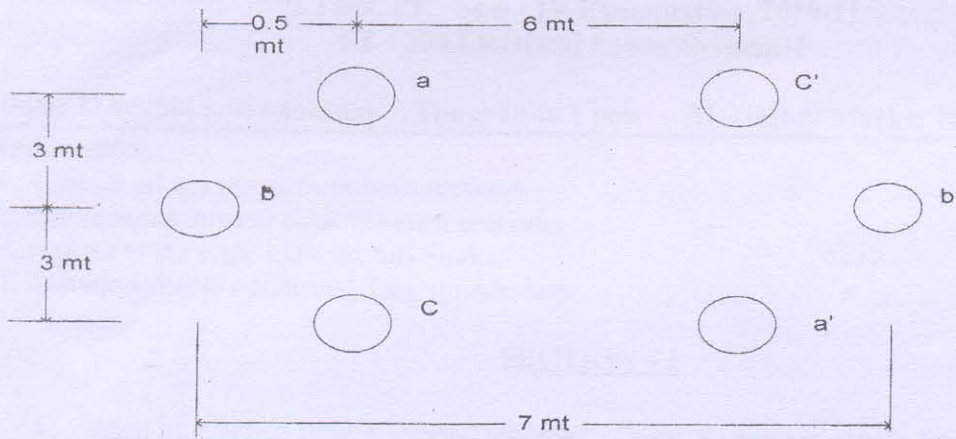


Fig. 1

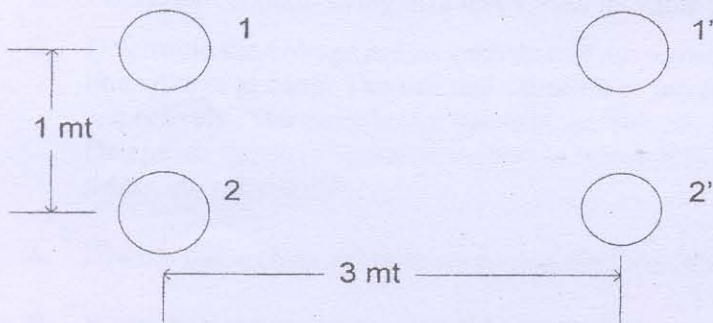


Fig. 2

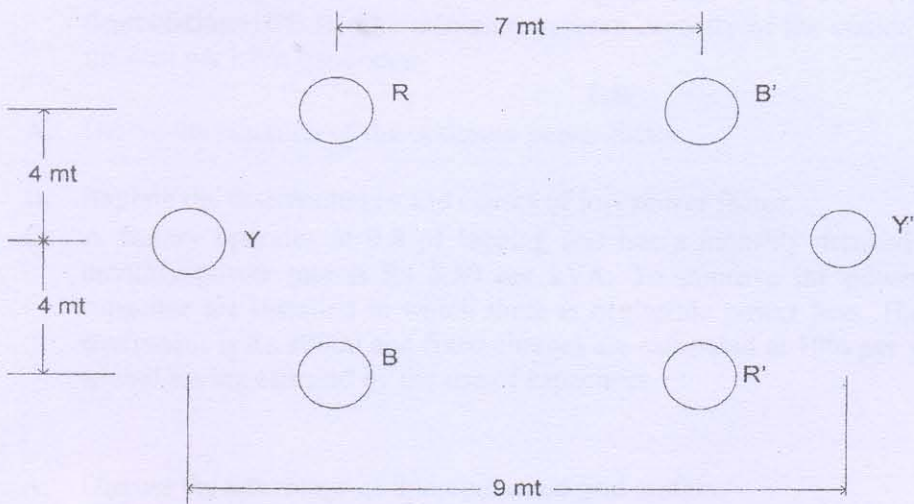


Fig. 3